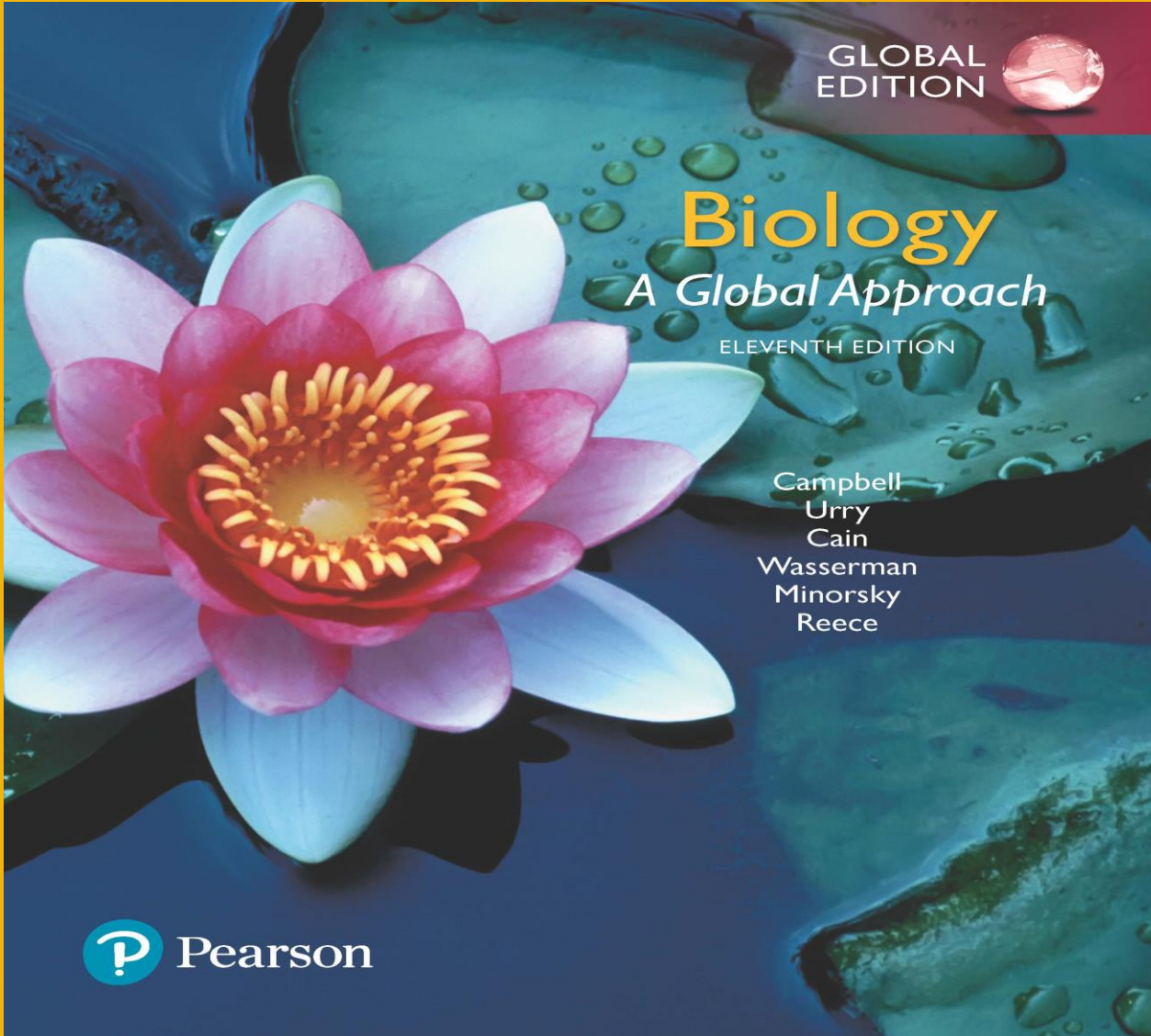




Chapter 20

The Evolution of Genomes

Lecture Presentations by
Nicole Tunbridge and
Kathleen Fitzpatrick

Reading the Leaves from the Tree of Life

- Complete genome sequences exist for a human, chimpanzee, *E. coli*, brewer's yeast, corn, fruit fly, elephant shark, and many other organisms
- Comparisons of genomes among organisms provide insights into evolution and other biological processes

- **Genomics** is the study of whole sets of genes and their interactions
- **Bioinformatics** is the application of computational methods to the storage and analysis of biological data

Concept 20.1: The Human Genome Project fostered development of faster, less expensive sequencing techniques

- Officially begun as the **Human Genome Project** in 1990, the sequencing of the human genome was largely completed in 2003
- The genome was completed using sequencing machines and the dideoxy chain termination method
- A major thrust of the project was development of technology for faster sequencing

- Two approaches complemented each other in obtaining the complete sequence
- The initial approach built on an earlier storehouse of human genetic information
- Then J. Craig Venter set up a company to sequence the entire genome using an alternative **whole-genome shotgun approach**
- This used cloning and sequencing of fragments of randomly cut DNA followed by assembly into a single continuous sequence

Figure 20.2_1

- 1 Cut the DNA into overlapping fragments short enough for sequencing.**
- 2 Clone the fragments in plasmid or other vectors.**

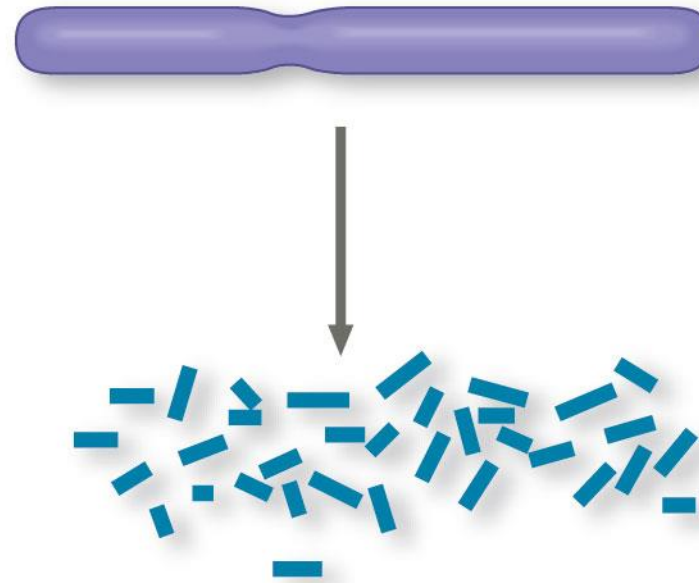


Figure 20.2_2

- 1 Cut the DNA into overlapping fragments short enough for sequencing.
- 2 Clone the fragments in plasmid or other vectors.
- 3 Sequence each fragment.

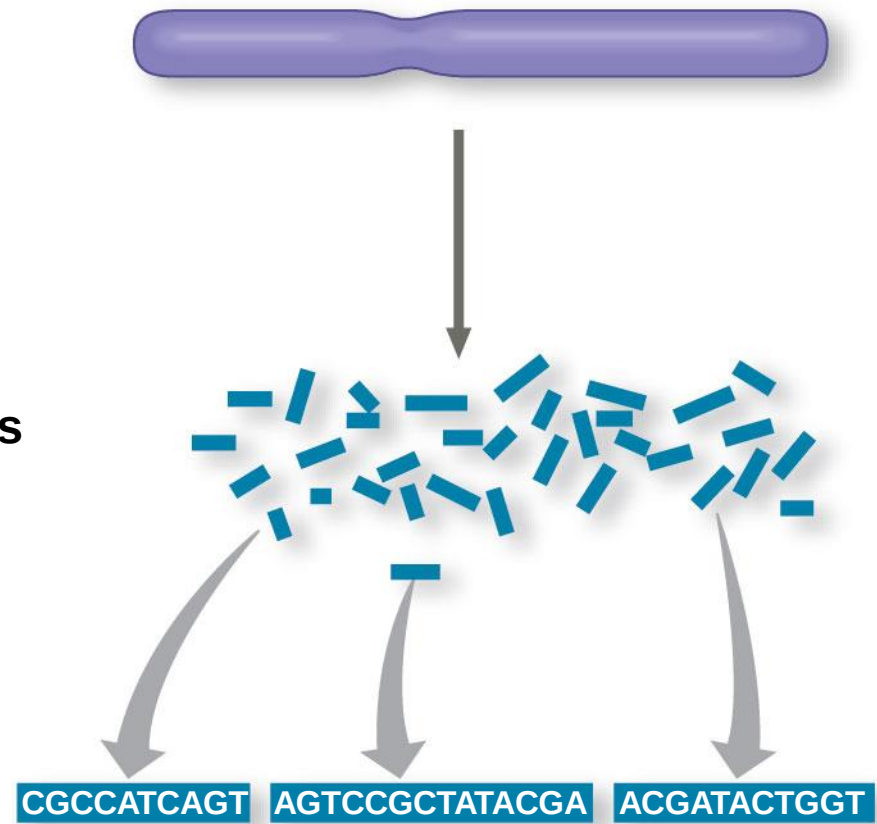
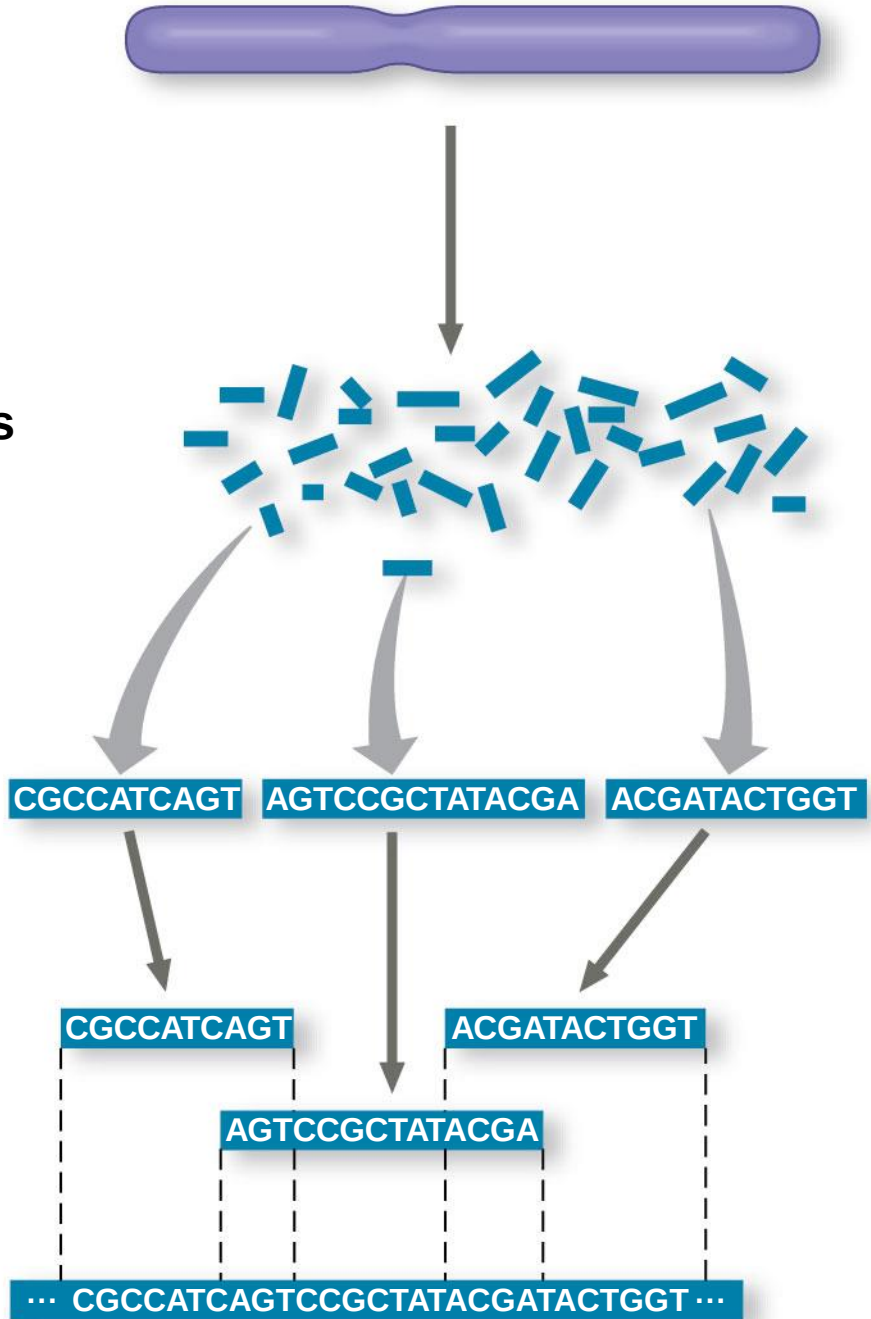


Figure 20.2_3

- 1 Cut the DNA into overlapping fragments short enough for sequencing.
- 2 Clone the fragments in plasmid or other vectors.
- 3 Sequence each fragment.
- 4 Order the sequences into one overall sequence with computer software.



- Today, the whole-genome shotgun approach is still used, though newer techniques are contributing to the faster pace and lowered cost of genome sequencing
- These “next-generation” techniques do not require a cloning step
- These techniques have also facilitated a **metagenomics** approach, in which DNA from a group of species in an environmental sample is sequenced

- Making sense of massive amounts of data from many genome sequences has necessitated new analytical approaches

Concept 20.2: Scientists use bioinformatics to analyze genomes and their functions

- The Human Genome Project established databases and refined analytical software to make data available on the Internet

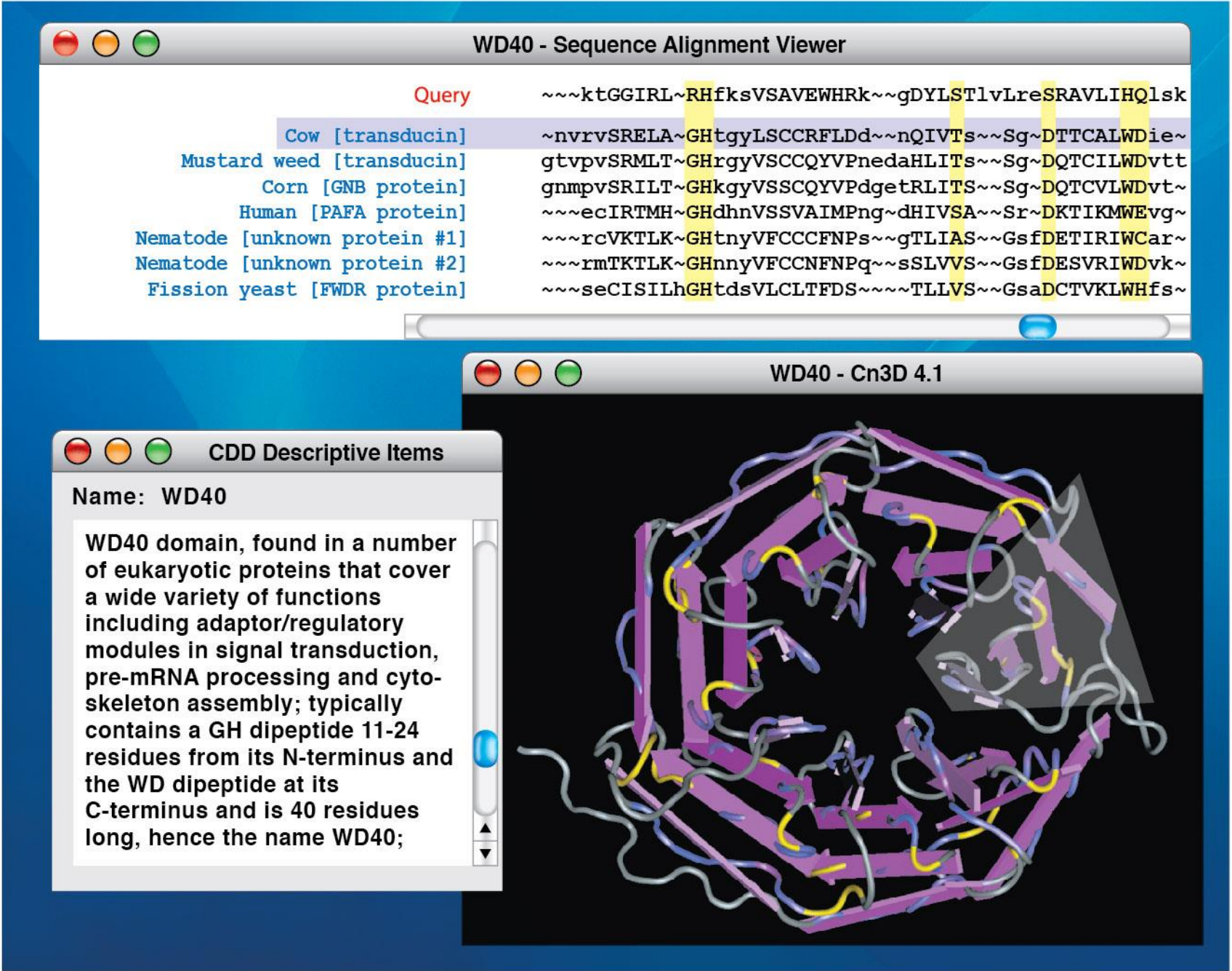
Centralized Resources for Analyzing Genome Sequences

- Bioinformatics resources are provided by a number of sources
 - The National Library of Medicine and the National Institutes of Health (NIH) created the National Center for Biotechnology Information (NCBI)
 - European Molecular Biology Laboratory
 - DNA Data Bank of Japan
 - BGI in Shenzhen, China

- GenBank, the NCBI database of sequences, is constantly updated, with the data it contains increasing rapidly
- The NCBI BLAST software allows users to search for
 - A specific DNA sequence
 - A predicted protein sequence
 - Common stretches of amino acids in a protein

- There is a software program that can align and compare a collection of sequences
- There is also a worldwide database of all three-dimensional protein structures
- The vast array of resources available to researchers around the world are free of charge

Figure 20.3



Identifying Protein-Coding Genes and Understanding Their Functions

- Using available DNA sequences, geneticists can study genes directly
- The identification of protein-coding genes within DNA sequences in a database is called **gene annotation**

- Gene annotation is largely an automated process
- Comparison of sequences of previously unknown genes with those of known genes in other species may help provide clues about their function
- The identities of these genes can be further confirmed by using RNA-seq or a similar method
- Functional studies involving knocking out the gene function can show how the phenotype is altered
- CRISPR-Cas9 can be used for such studies

Understanding Genes and Gene Expression at the Systems Level

- Genomics is a rich source of new insights into questions about genome organization, regulation of gene expression, embryonic development, and evolution
- The ENCODE (Encyclopedia of DNA Elements) project ran from 2003 to 2012
- The aim was to learn about the functionally important elements in the human genome

- Besides working to identify enhancers and promoters, investigators also extensively characterized histone and DNA modifications and chromatin structure
- This project allows comparison of results from different projects, yielding a richer picture of the whole genome
- About 75% of the genome is transcribed at some point in at least one cell type studied

- A related project called the Roadmap Epigenomics Project set out to characterize the epigenetic features of the genome (the epigenome)
- A useful finding was that the original tissue in which a cancer arose can be identified in a secondary tumor based on its epigenomic features

Systems Biology

- A **proteome** is the entire set of proteins expressed by a cell or group of cells
- **Proteomics** is an approach to studying large sets of proteins and their properties
- Scientist have begun to compile catalogs of genes and proteins and have begun to focus on their functional integration in biological systems
- This approach is called **systems biology**

- Researchers working on the yeast *Saccharomyces cerevisiae* used sophisticated techniques to disable pairs of genes one pair at a time, creating double mutants
- Computer software then mapped genes to produce a network-like “functional map” of their interactions
- The systems biology approach is possible because of advances in bioinformatics

Figure 20.4

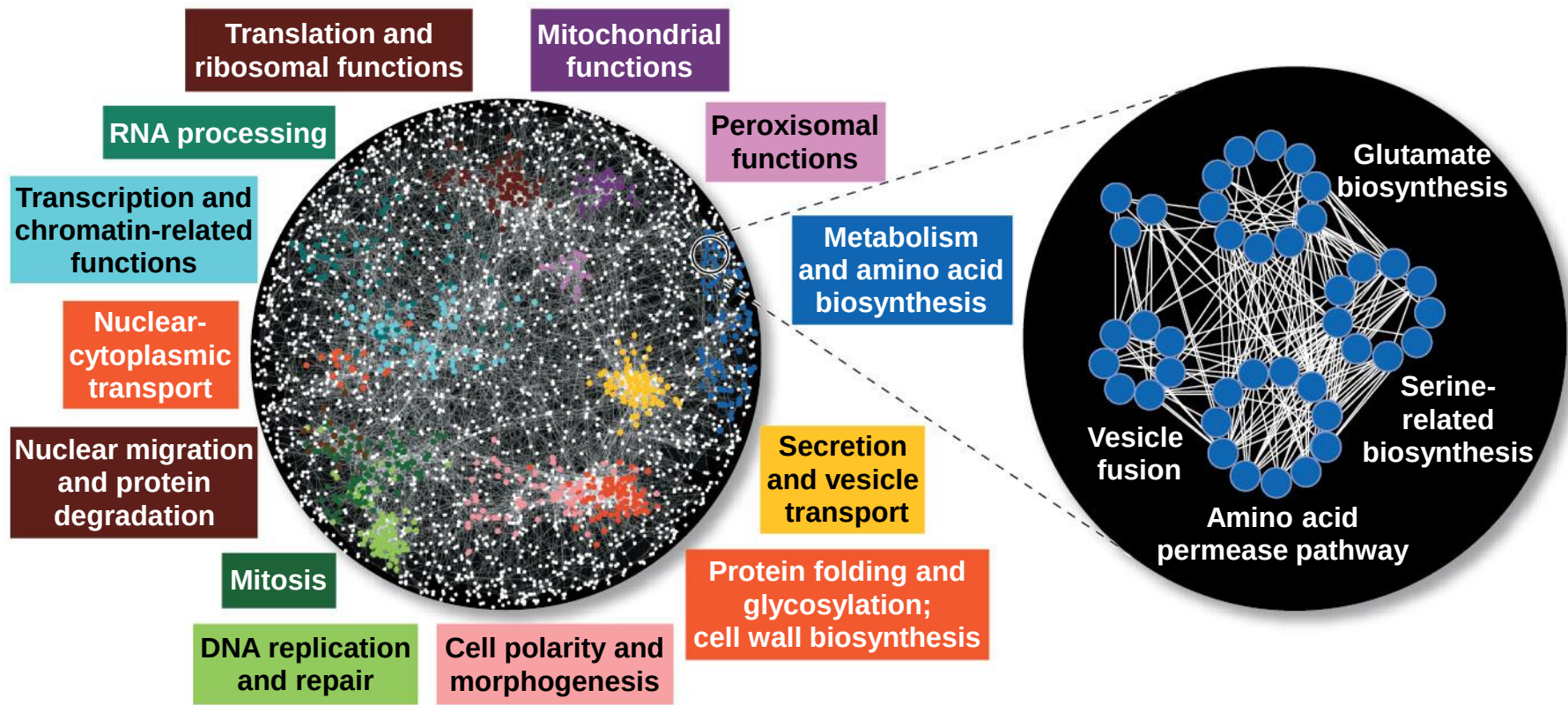


Figure 20.4a

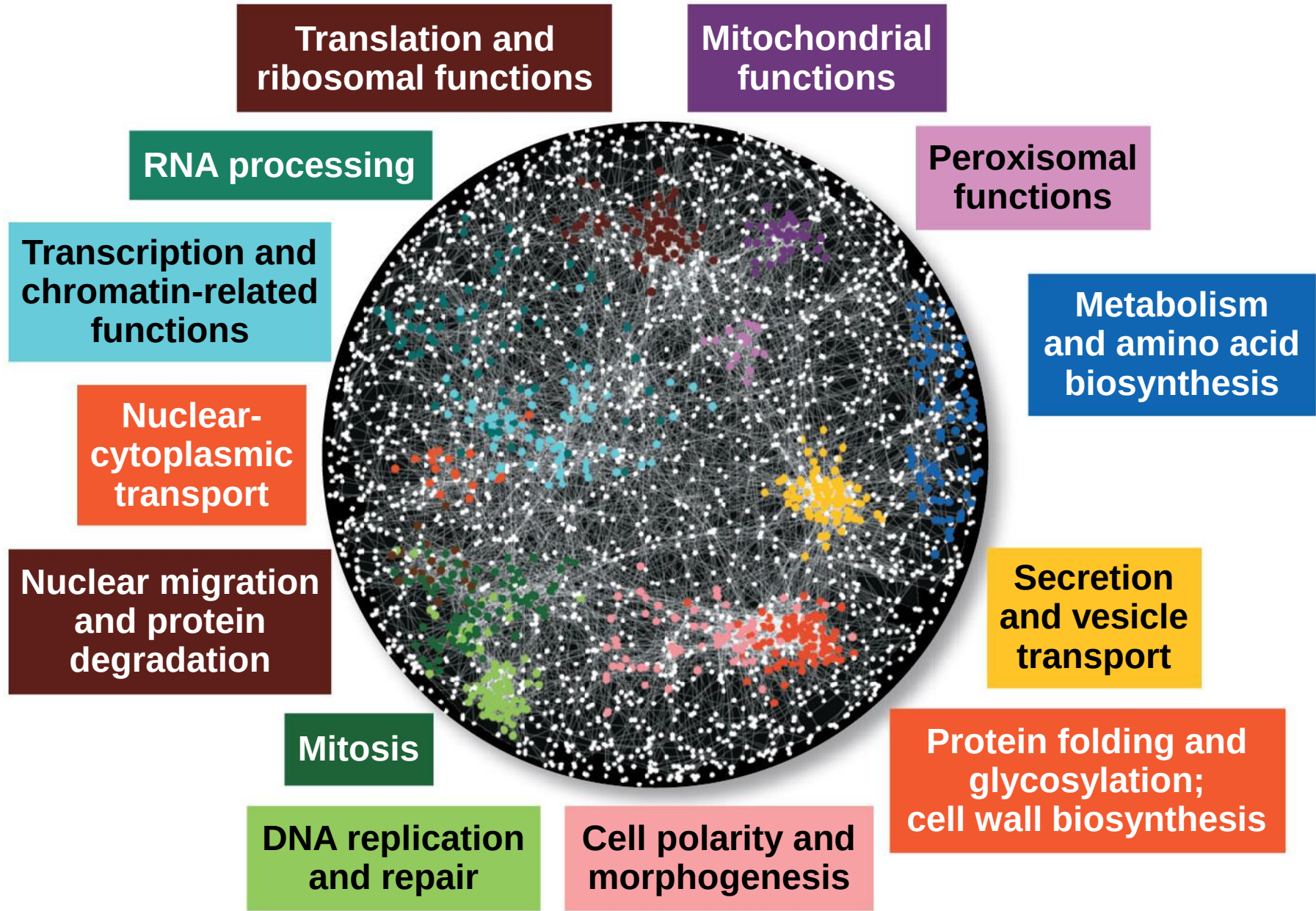
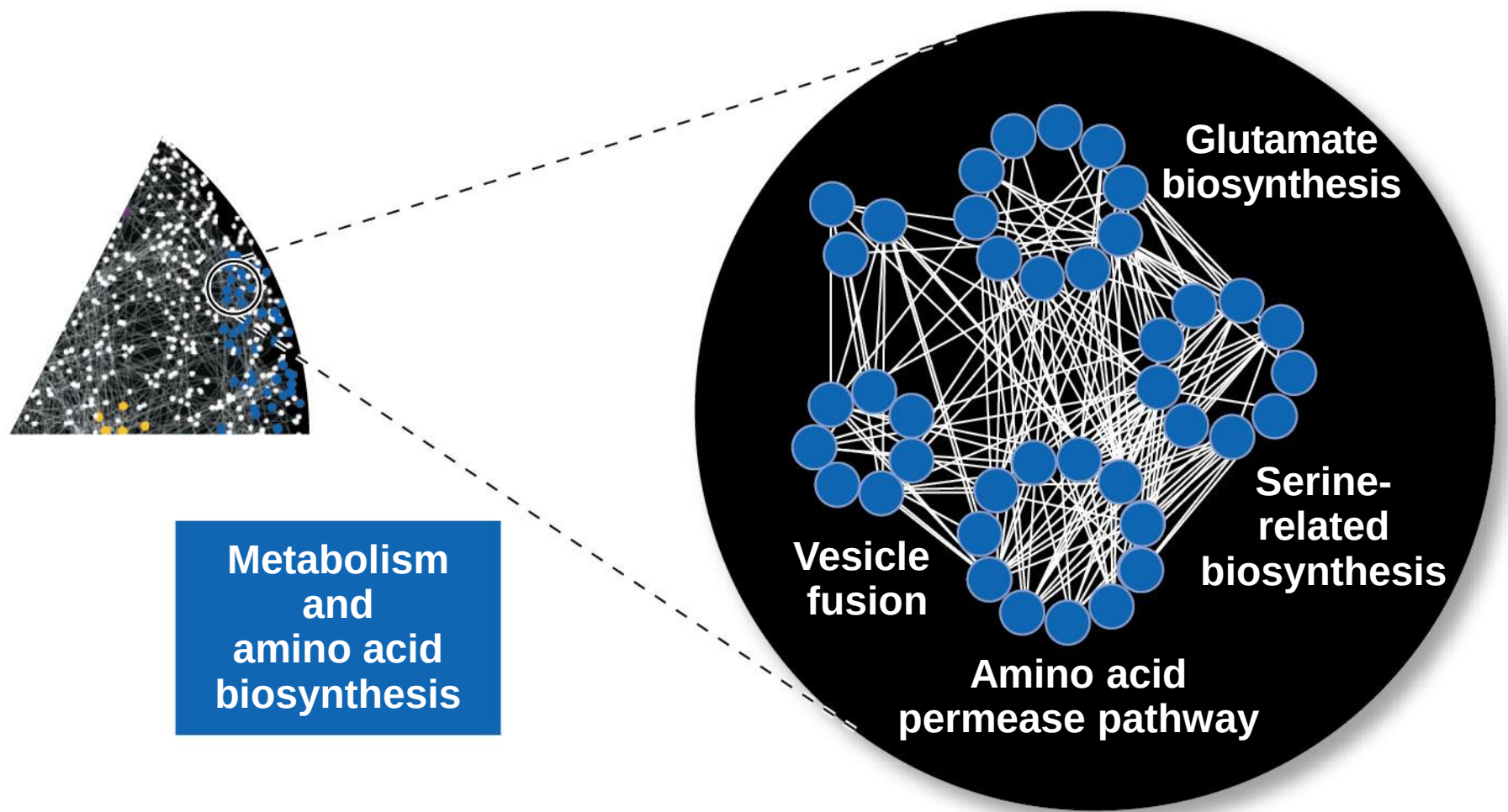


Figure 20.4b



Application of Systems Biology to Medicine

- The Cancer Genome Atlas project, completed in 2010, looked for all the common mutations in three types of cancer by comparing gene sequences and expression in cancer versus normal cells
- This was so fruitful that it has been extended to ten other common cancers
- Silicon and glass “chips” have been produced that hold a microarray of most known human genes
- These are used to study gene expression patterns in patients suffering from various cancers or other diseases

Figure 20.5



Concept 20.3: Genomes vary in size, number of genes, and gene density

- The sequences of thousands of genomes have been completed
- Tens of thousands of genomes are either in progress or are considered permanent drafts

Genome Size

- Genomes of most bacteria and archaea range from 1 to 6 million base pairs (Mb)
- Genomes of archaea are mostly within this size range
- Eukaryotic genomes tend to be larger
- Most plants and animals have genomes greater than 100 Mb; humans have 3,000 Mb
- Within each domain there is no systematic relationship between genome size and phenotype

Table 20.1

Table 20.1 Genome Sizes and Estimated Numbers of Genes*

Organism	Haploid Genome Size (Mb)	Number of Genes	Genes per Mb
<u>Bacteria</u>			
<i>Haemophilus influenzae</i>	1.8	1,700	940
<i>Escherichia coli</i>	4.6	4,400	950
<u>Archaea</u>			
<i>Archaeoglobus fulgidus</i>	2.2	2,500	1,130
<i>Methanosarcina barkeri</i>	4.8	3,600	750
<u>Eukaryotes</u>			
<i>Saccharomyces cerevisiae</i> (yeast, a fungus)	12	6,300	525
<i>Urticularia gibba</i> (floating bladderwort)	82	28,500	348
<i>Caenorhabditis elegans</i> (nematode)	100	20,100	200
<i>Arabidopsis thaliana</i> (mustard family plant)	120	27,000	225
<i>Drosophila melanogaster</i> (fruit fly)	165	14,000	85
<i>Daphnia pulex</i> (water flea)	200	31,000	155
<i>Zea mays</i> (corn)	2,300	32,000	14
<i>Ailuropoda melanoleuca</i> (giant panda)	2,400	21,000	9
<i>Homo sapiens</i> (human)	3,000	< 21,000	7
<i>Paris japonica</i> (Japanese canopy plant)	149,000	ND	ND
*Some values given here are likely to be revised as genome analysis continues. Mb = million base pairs. ND = not determined.			

Number of Genes

- Free-living bacteria and archaea have 1,500 to 7,500 genes
- Unicellular fungi have about 5,000 genes and multicellular eukaryotes up to at least 40,000 genes
- Number of genes is not correlated to genome size

- It is estimated that the nematode *C. elegans* has 100 Mb and 20,100 genes, while *Drosophila* has 165 Mb and 14,000 genes
- Researchers predicted the human genome would contain about 50,000 to 100,000 genes; however, the number is fewer than 21,000
- Vertebrate genomes can produce more than one polypeptide per gene because of alternative splicing of RNA transcripts

Gene Density and Noncoding DNA

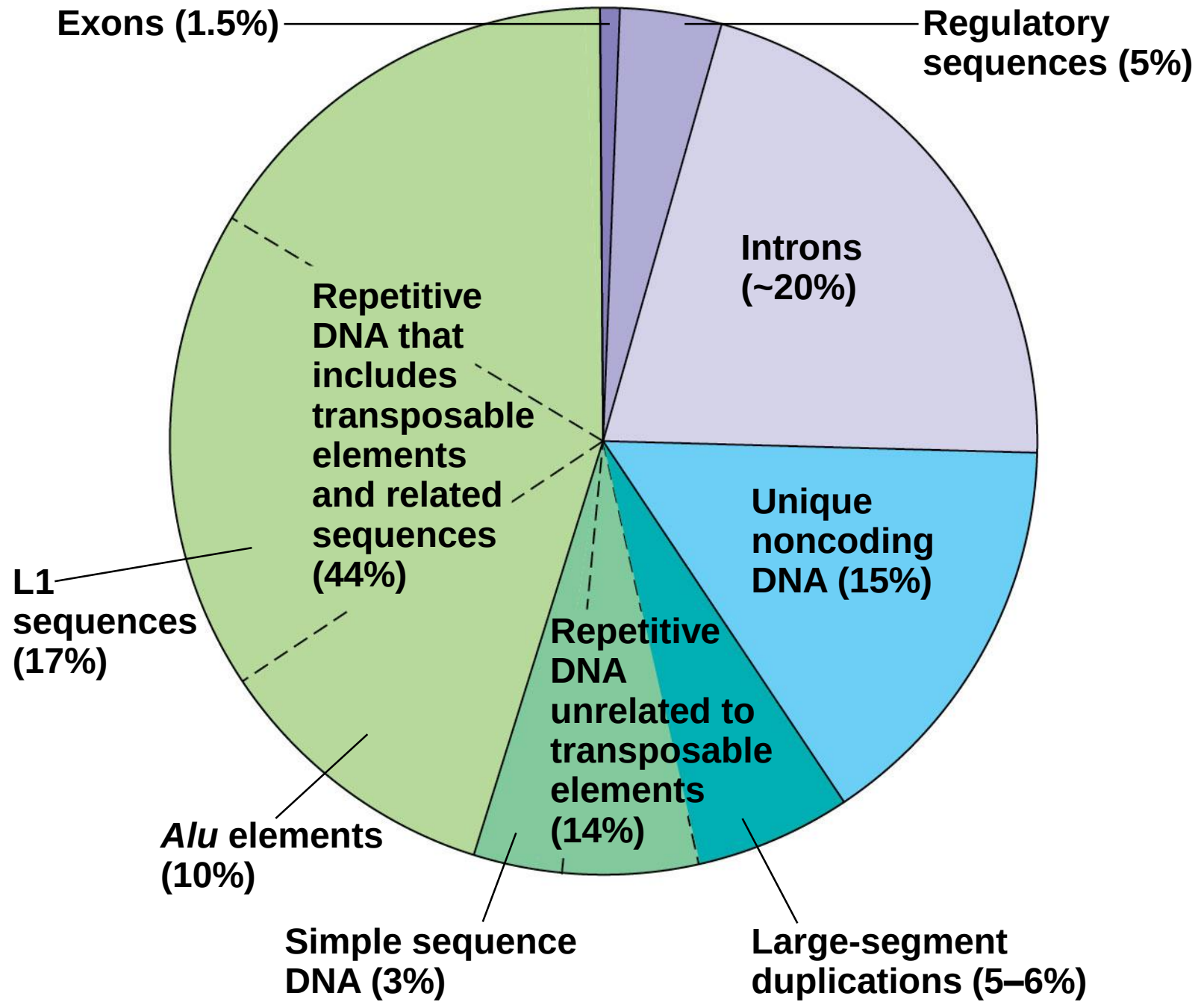
- Humans and other mammals have the lowest gene density, or number of genes, in a given length of DNA
- Multicellular eukaryotes have many introns within genes and a large amount of noncoding DNA between genes

Concept 20.4: Multicellular eukaryotes have a lot of noncoding DNA and many multigene families

- Sequencing of the human genome revealed that 98.5% does not code for proteins, rRNAs, or tRNAs
- Gene regulatory sequences and introns account for 5% and 20%, respectively, of the human genome

- Noncoding DNA found between genes includes
 - **Pseudogenes**, former genes that have accumulated mutations and are nonfunctional
 - **Repetitive DNA**, present in multiple copies in the genome
- A high level of sequence conservation in some noncoding DNA among humans, rats, and mice suggests that these regions have important functions

Figure 20.6



Transposable Elements and Related Sequences

- Prokaryotes and eukaryotes have stretches of DNA that can move from one location to another within the genome, called **transposable elements**
- About 75% of human repetitive DNA is made up of transposable elements and the sequences related to them
- The first evidence of these mobile elements came from Barbara McClintock's breeding experiments with Indian corn (maize)

Figure 20.7



Figure 20.7a



Figure 20.7b



Movement of Transposons and Retrotransposons

- Eukaryotic transposable elements are of two types
 - **Transposons**, which move by means of a DNA intermediate and require a transposase enzyme
 - **Retrotransposons**, which move by means of an RNA intermediate using a reverse transcriptase

Figure 20.8

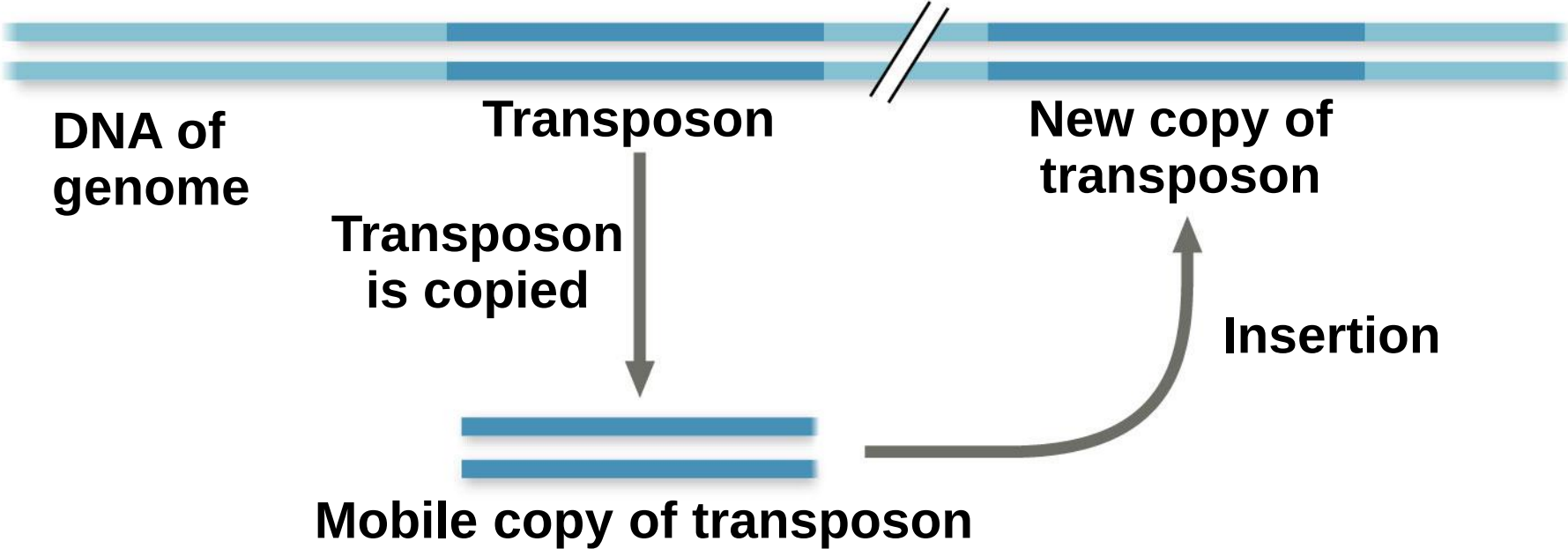
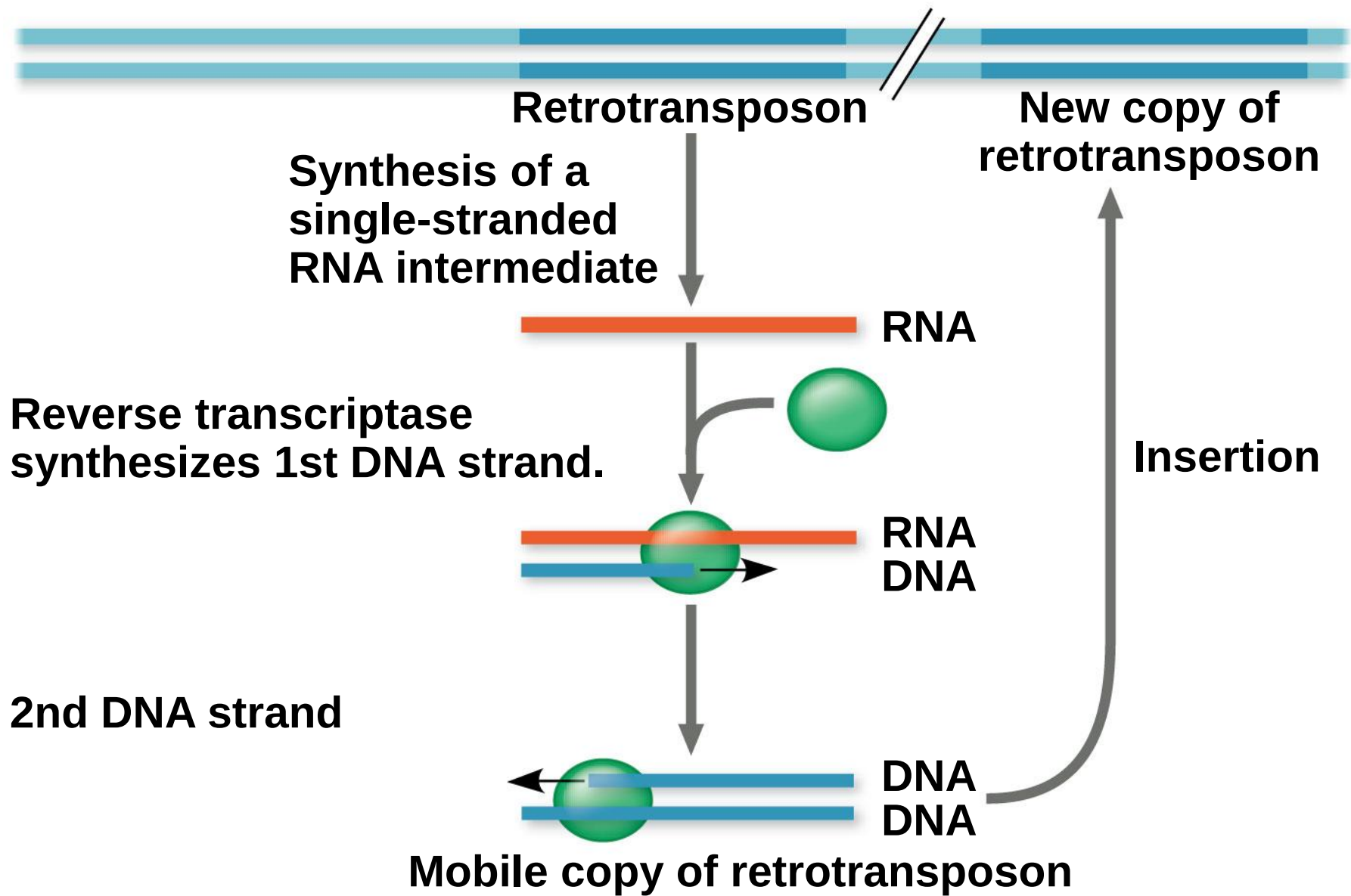


Figure 20.9



Sequences Related to Transposable Elements

- Multiple copies of transposable elements and related sequences are scattered throughout eukaryotic genomes
- In primates, a large portion of transposable element–related DNA consists of a family of similar sequences called *Alu* elements
- Many *Alu* elements are transcribed into RNA molecules; some are thought to help regulate gene expression

- The human genome also contains many sequences of a type of retrotransposon called LINE-1 (L1)
- L1 sequences have a low rate of transposition and may have effects on gene expression
- L1 transposons may play roles in the diversity of neuronal cell types

Other Repetitive DNA, Including Simple Sequence DNA

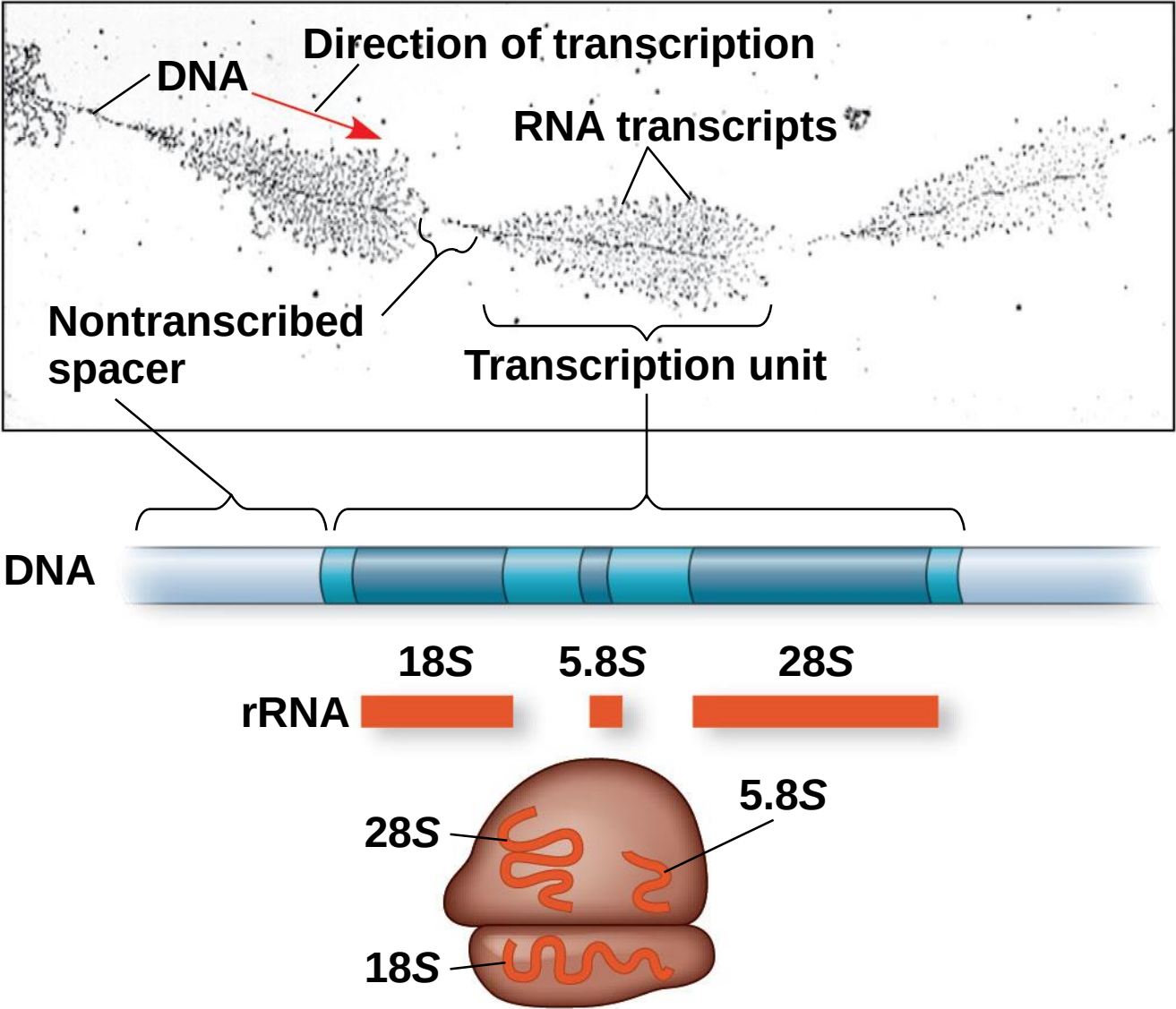
- About 5–6% of the human genome consists of duplication of long sequences of DNA from one location to another
- In contrast, **simple sequence DNA** contains many copies of tandemly repeated short sequences

- A series of repeating units of 2 to 5 nucleotides is called a **short tandem repeat (STR)**
- The repeat number for STRs can vary among sites (within a genome) or individuals
- Simple sequence DNA is common in centromeres and telomeres, where it probably plays structural roles in the chromosome

Genes and Multigene Families

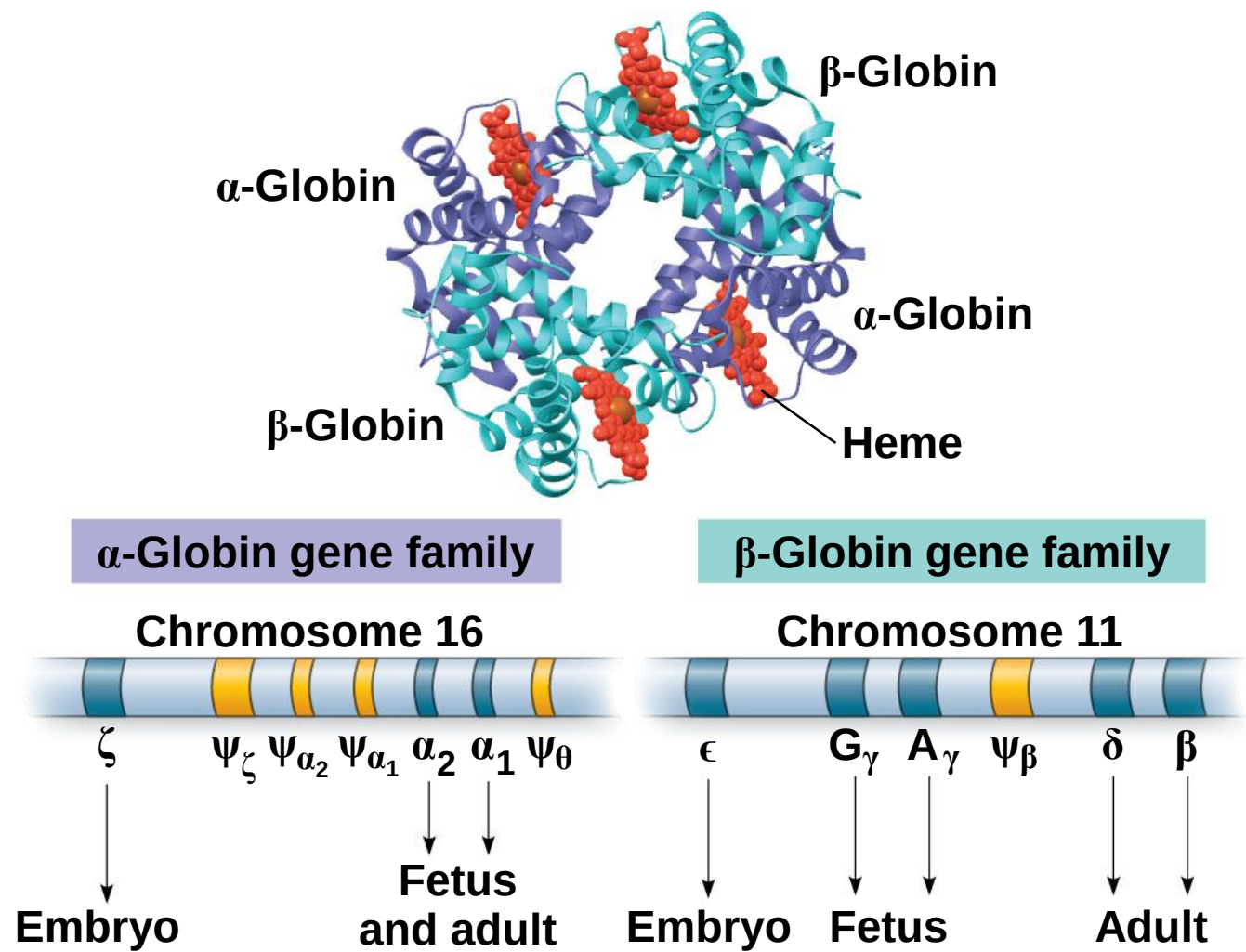
- Many eukaryotic genes are present in one copy per haploid set of chromosomes
- The rest of the genes occur in **multigene families**, collections of two or more identical or very similar genes
- Some multigene families consist of identical DNA sequences, usually clustered tandemly, such as those that code for rRNA products

Figure 20.10a



(a) Part of the ribosomal RNA gene family

- The classic examples of multigene families of nonidentical genes are two related families of genes that encode globins
- α -globins and β -globins are polypeptides of hemoglobin coded by genes on different human chromosomes and are expressed at different times in development



(b) The human α -globin and β -globin gene families

Concept 20.5: Duplication, rearrangement, and mutation of DNA contribute to genome evolution

- The basis of change at the genomic level is mutation, which underlies much of genome evolution
- The earliest forms of life likely had only those genes necessary for survival and reproduction
- The size of genomes has increased over evolutionary time, with the extra genetic material providing raw material for gene diversification

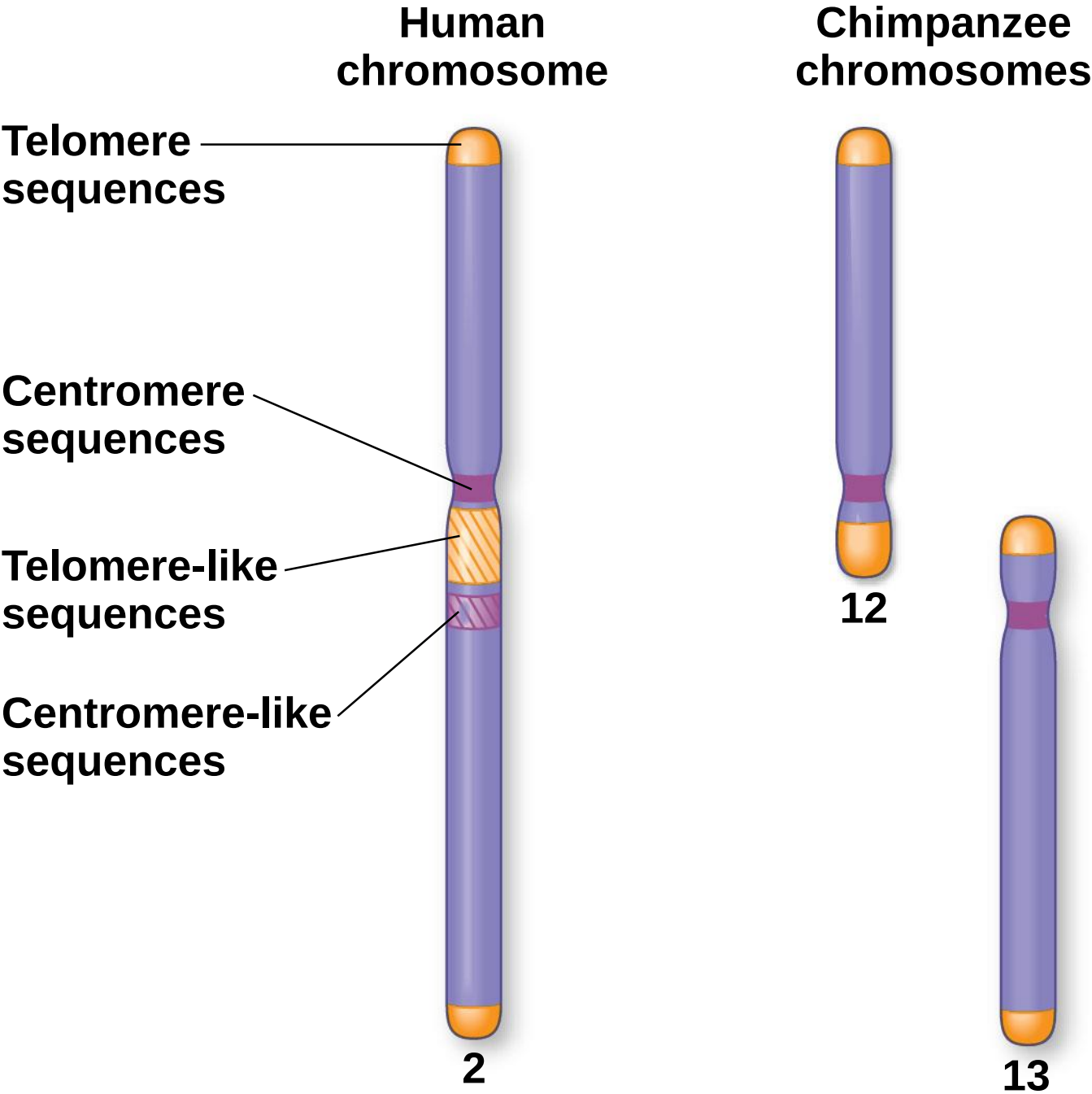
Duplication of Entire Chromosome Sets

- Accidents in meiosis can lead to one or more extra sets of chromosomes, a condition known as polyploidy
- The genes in one or more of the extra sets can diverge by accumulating mutations
- These variations may persist if the organism carrying them survives and reproduces
- In this way, genes with novel functions can evolve

Alterations of Chromosome Structure

- Humans have 23 pairs of chromosomes, while chimpanzees have 24 pairs
- Following the divergence of humans and chimpanzees from a common ancestor, two ancestral chromosomes fused in the human line
- Large blocks of genes on human chromosome 16 are found on four mouse chromosomes
- This indicates that the genes in each block stayed together in both the human and mouse lineages

Figure 20.11



Human chromosome



16

Mouse chromosomes



7



8



16



17

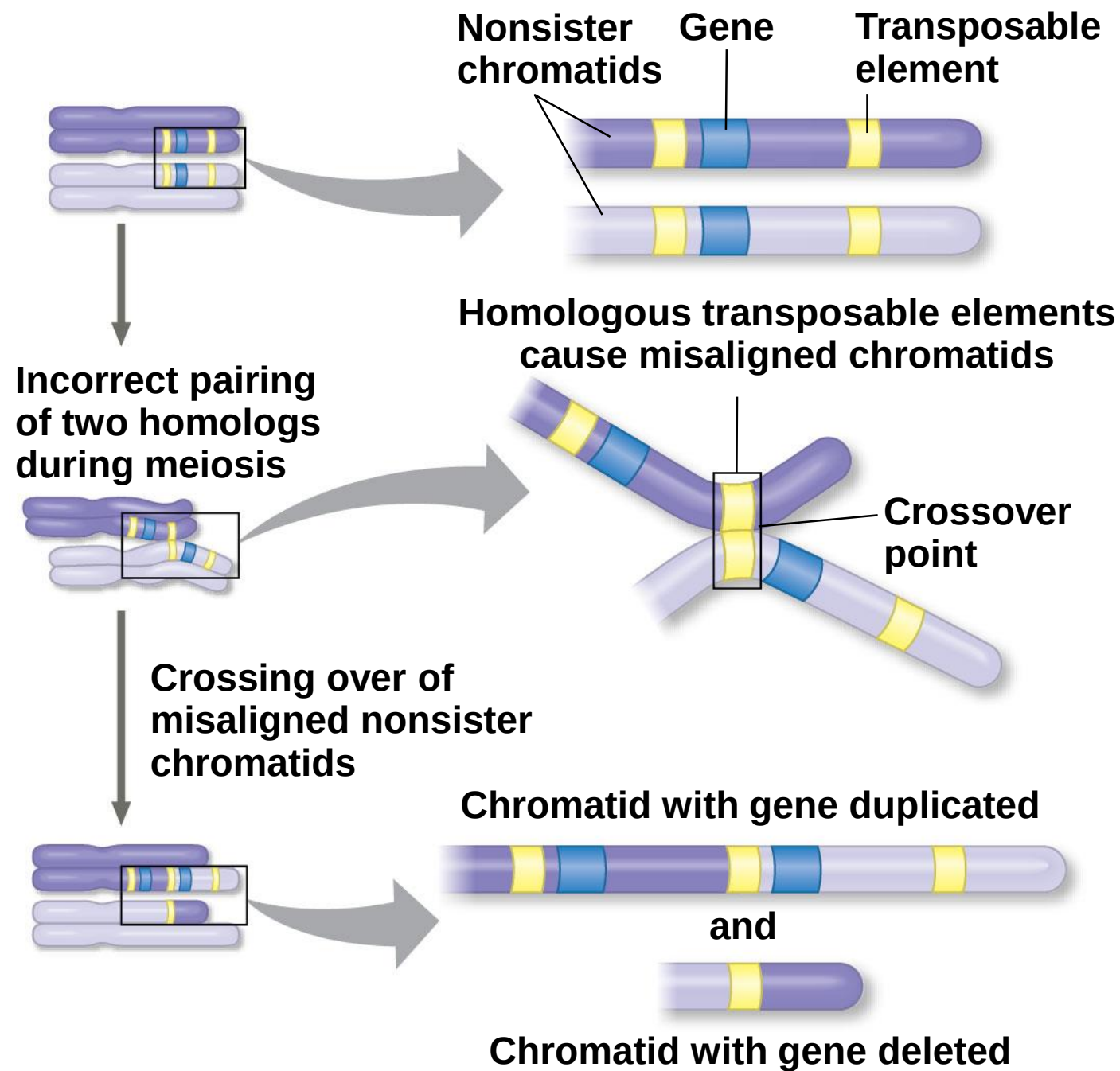
- Duplications and inversions result from mistakes during meiotic recombination
- Comparative analysis between chromosomes of humans and seven mammalian species paints a hypothetical chromosomal evolutionary history

- The rate of duplications and inversions seems to have accelerated about 100 million years ago
- This coincides with when large dinosaurs went extinct and mammals diversified
- Chromosomal rearrangements are thought to contribute to the generation of new species

Duplication and Divergence of Gene-Sized Regions of DNA

- Unequal crossing over during prophase I of meiosis can result in one chromosome with a deletion and another with a duplication of a particular region
- Transposable elements can provide sites for crossover between nonsister chromatids

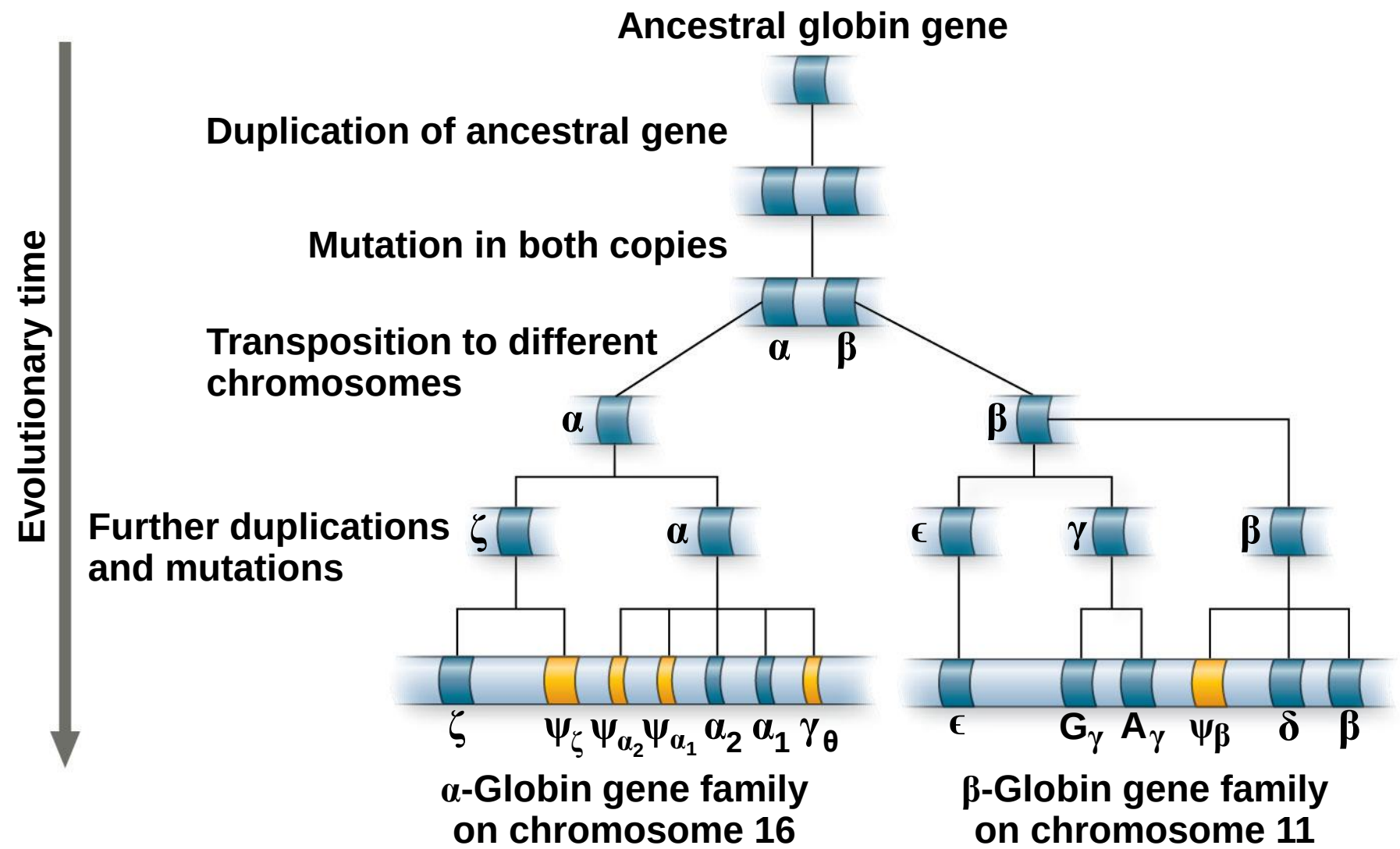
Figure 20.13



Evolution of Genes with Related Functions: The Human Globin Genes

- The genes encoding the various globin proteins evolved from one common ancestral globin gene, which duplicated and diverged about 450–500 million years ago
- After the duplication events, differences between the genes in the globin family arose from the accumulation of mutations

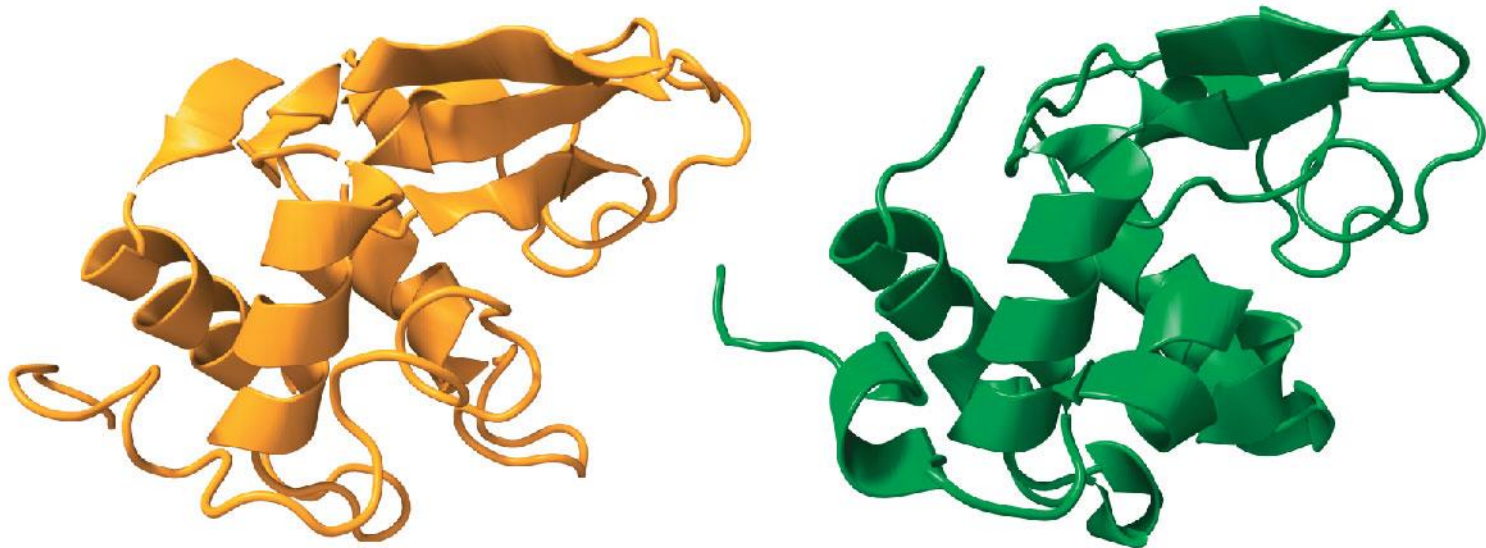
Figure 20.14



- Subsequent duplications of these genes and random mutations gave rise to the present globin genes, which code for oxygen-binding proteins
- The similarity in the amino acid sequences of the various globin proteins supports this model of gene duplication and mutation

Evolution of Genes with Novel Functions

- The copies of some duplicated genes have diverged so much in evolution that the functions of their encoded proteins are now very different
- For example the lysozyme gene was duplicated and evolved into the gene that encodes α -lactalbumin in mammals
- Lysozyme is an enzyme that helps protect animals against bacterial infection
- α -lactalbumin is a nonenzymatic protein that plays a role in milk production in mammals



(a) Lysozyme

(b) α -lactalbumin

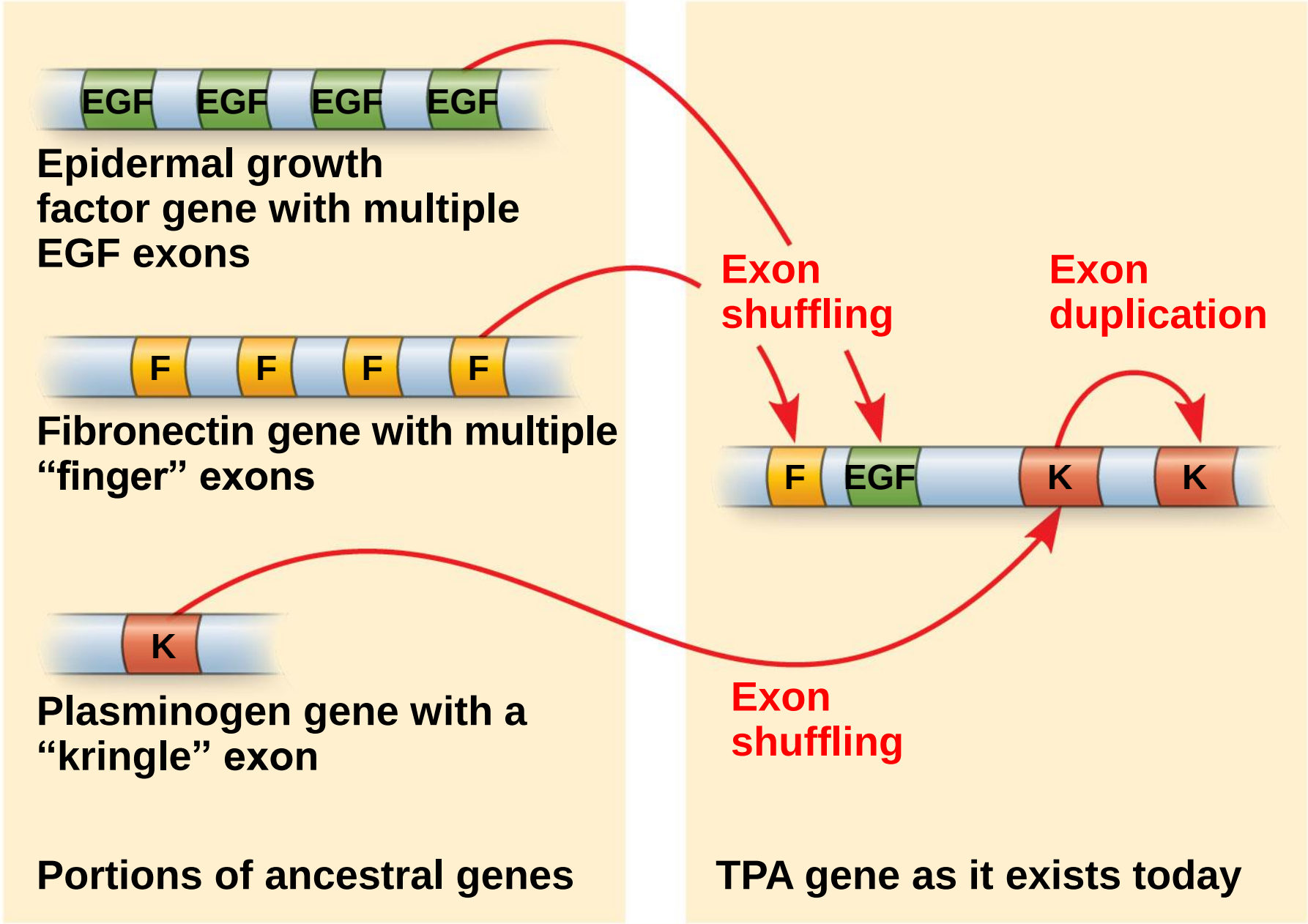
Lysozyme	1	KVFERCELAR	TLKRLGMDGY	RGISLANWMC	LAKWESGYNT	RATNYNAGDR
α -lactalbumin	1	KQFTKCELSQ	LLK--DIDGY	GGIALPELIC	TMFHTSGYDT	QAIVENN--E
Lysozyme	51	STDYGIFQIN	SRYWCNDGKT	PGAVNACHLS	CSALLQDNIA	DAVACAQRVV
α -lactalbumin	51	STEYGLFQIS	NKLWCKSSQV	PQSRNICDIS	CDKFLDDIT	DDIMCAKKIL
Lysozyme	101	RDPQGIRAWV	AWRNRCQ-NR	DVRQYVQGCG	V	
α -lactalbumin	101	D-IKGIDYWL	AHKALCT--E	KLEQWLCEKL	-	

(c) Amino acid sequence alignments of lysozyme and α -lactalbumin

Rearrangements of Parts of Genes: Exon Duplication and Exon Shuffling

- Errors in meiosis can result in an exon being duplicated on one chromosome and deleted from the homologous chromosome
- In exon shuffling, errors in meiotic recombination lead to some mixing and matching of exons, either within a gene or between two nonallelic genes
- The current version of the gene for tissue plasminogen activator (TPA) is thought to have arisen by several instances of exon shuffling and subsequent duplication

Figure 20.16



How Transposable Elements Contribute to Genome Evolution

- Multiple copies of similar transposable elements facilitate recombination, or crossing over, between different chromosomes
- Insertion of transposable elements within a protein-coding sequence may block protein production
- Insertion of transposable elements within a regulatory sequence may increase or decrease protein production

- Transposable elements may carry a gene or groups of genes to a new position
- Transposable elements may also create new sites for alternative splicing in an RNA transcript
- In all cases, changes are usually detrimental but may on occasion prove advantageous to an organism